# LLD & HLD MCQ Question Bank

60 questions | Difficulty 0.5 to 10 (step 0.5, 3 questions per level) | 4 options each

**Q1. [Difficulty: 0.5] What does LLD stand for in software design?**

A) Linear Logic Design
B) Logical Layer Design
C) Long-Level Design
D) Low-Level Design

Answer: D

**Q2. [Difficulty: 0.5] What does HLD stand for in software design?**

A) High-Level Design
B) Hardware Level Design
C) Hybrid Level Design
D) Heavy Load Design

Answer: A

**Q3. [Difficulty: 0.5] Which of the following is a relational database?**

A) Cassandra
B) MongoDB
C) Redis
D) MySQL

Answer: D

**Q4. [Difficulty: 1] What is a server in a client-server architecture?**

A) A machine or program that receives client requests and sends back responses
B) A device used only for storing backups
C) A cable that connects two computers
D) A program that displays web pages to the user

Answer: A

**Q5. [Difficulty: 1] What does API stand for?**

A) Applied Program Integration
B) Application Process Instruction
C) Automated Process Interface
D) Application Programming Interface

Answer: D

**Q6. [Difficulty: 1] Which of the following is a NoSQL database?**

A) MySQL
B) MongoDB
C) PostgreSQL
D) Oracle

Answer: B

**Q7. [Difficulty: 1.5] What does a load balancer do?**

A) Distributes incoming traffic across multiple servers
B) Stores data permanently
C) Compiles source code
D) Encrypts all database records

Answer: A

**Q8. [Difficulty: 1.5] What is caching?**

A) Compressing all files on a server
B) Copying data to tape backups
C) Deleting old data from a database
D) Storing frequently accessed data in a faster storage layer

Answer: D

**Q9. [Difficulty: 1.5] Which design pattern ensures a class has only one instance?**

A) Singleton
B) Decorator
C) Factory
D) Observer

Answer: A

**Q10. [Difficulty: 2] Which data format is most commonly used in REST APIs?**

A) BMP
B) MP3
C) EXE
D) JSON

Answer: D

**Q11. [Difficulty: 2] Which HTTP method is normally used to retrieve data?**

A) DELETE
B) GET
C) PUT
D) POST

Answer: B

**Q12. [Difficulty: 2] What does a primary key do in a database table?**

A) Uniquely identifies each row
B) Links the table to a file
C) Sorts the table alphabetically
D) Encrypts the table

Answer: A

**Q13. [Difficulty: 2.5] What is horizontal scaling?**

A) Adding more CPU and RAM to one machine
B) Adding more machines to handle load
C) Reducing the number of servers
D) Moving a server to another country

Answer: B

**Q14. [Difficulty: 2.5] Which design pattern creates objects without exposing the creation logic to the client?**

A) Factory
B) Adapter
C) Observer
D) Iterator

Answer: A

**Q15. [Difficulty: 2.5] What does an SQL JOIN do?**

A) Deletes duplicate rows
B) Creates a new database
C) Encrypts two tables together
D) Combines rows from two or more tables using a related column

Answer: D

**Q16. [Difficulty: 3] What is a database index used for?**

A) Backing up the table
B) Limiting the number of rows
C) Encrypting the column
D) Speeding up data lookups

Answer: D

**Q17. [Difficulty: 3] What does the Observer pattern do?**

A) Notifies dependent objects automatically when an object's state changes
B) Adds new behaviour to objects by wrapping them
C) Creates objects without specifying their class
D) Ensures only one object exists

Answer: A

**Q18. [Difficulty: 3] What is a CDN?**

A) A network of distributed servers that delivers content from locations close to users
B) A type of message queue
C) A tool for compiling code
D) A database that stores user passwords

Answer: A

**Q19. [Difficulty: 3.5] Which type of database is usually preferred when strict schemas and multi-row transactions are needed?**

A) In-memory cache
B) Relational (SQL) database
C) Graph database only
D) Key-value store

Answer: B

**Q20. [Difficulty: 3.5] What does it mean for a server to be stateless?**

A) It stores no data at all
B) It can handle only one request
C) It does not store client session data between requests
D) It has no operating system

Answer: C

**Q21. [Difficulty: 3.5] What is a message queue mainly used for?**

A) Rendering web pages
B) Compressing images
C) Storing user passwords
D) Asynchronous communication between services

Answer: D

**Q22. [Difficulty: 4] What is the Strategy pattern?**

A) It builds objects step by step
B) It ensures only one instance of a class exists
C) It lets you swap interchangeable algorithms at run time by encapsulating each one in a class
D) It converts one interface into another

Answer: C

**Q23. [Difficulty: 4] What does ACID stand for in databases?**

A) Atomicity, Concurrency, Isolation, Distribution
B) Accuracy, Consistency, Integrity, Durability
C) Atomicity, Consistency, Isolation, Durability
D) Availability, Consistency, Isolation, Durability

Answer: C

**Q24. [Difficulty: 4] What does CAP stand for in the CAP theorem?**

A) Consistency, Atomicity, Persistence
B) Consistency, Availability, Partition tolerance
C) Concurrency, Availability, Partitioning
D) Caching, Availability, Performance

Answer: B

**Q25. [Difficulty: 4.5] Which combination of data structures gives an O(1) LRU cache?**

A) A heap and a hash set
B) An array and a stack
C) A binary search tree and a queue
D) A hash map and a doubly linked list

Answer: D

**Q26. [Difficulty: 4.5] What is database sharding?**

A) Compressing tables to save space
B) Copying the same data to multiple servers
C) Encrypting data with several keys
D) Splitting data across multiple databases using a shard key

Answer: D

**Q27. [Difficulty: 4.5] What is a reverse proxy?**

A) A server that sits in front of backend servers and forwards client requests to them
B) A database replica
C) A server that hides the client's identity when browsing
D) A tool that reverses the order of packets

Answer: A

**Q28. [Difficulty: 5] What does the Decorator pattern do?**

A) Creates a family of related objects
B) Defines a one-to-many dependency
C) Adds behaviour to an object dynamically by wrapping it
D) Provides a single global access point

Answer: C

**Q29. [Difficulty: 5] What is the difference between replication and sharding?**

A) Replication splits data; sharding copies the same data
B) Replication copies the same data to several nodes; sharding splits data across nodes
C) Replication is used only for caches
D) Both split data in exactly the same way

Answer: B

**Q30. [Difficulty: 5] Which cache write strategy writes to the cache and the database at the same time?**

A) Write-around
B) Write-back
C) Write-through
D) Read-through

Answer: C

**Q31. [Difficulty: 5.5] Which rate-limiting algorithm allows short bursts of traffic up to a bucket size?**

A) Round robin
B) Leaky bucket
C) Token bucket
D) Fixed window counter

Answer: C

**Q32. [Difficulty: 5.5] What does the Adapter pattern do?**

A) Ensures a class has one instance
B) Notifies observers of changes
C) Creates objects without exposing creation logic
D) Converts one interface into another that the client expects

Answer: D

**Q33. [Difficulty: 5.5] What is eventual consistency?**

A) Writes are rejected until all nodes agree
B) Replicas may differ temporarily but converge if no new updates arrive
C) Data is never replicated
D) All replicas are always identical

Answer: B

**Q34. [Difficulty: 6] What is the main purpose of consistent hashing?**

A) To encrypt keys before storing them
B) To sort keys alphabetically
C) To minimize data movement when nodes are added to or removed from a cluster
D) To make all hash values unique

Answer: C

**Q35. [Difficulty: 6] How does the cache-aside pattern work?**

A) The database pushes every update to the cache
B) The application checks the cache first, and on a miss reads from the database and fills the cache
C) The cache itself loads data from the database on a miss
D) Every write goes only to the cache

Answer: B

**Q36. [Difficulty: 6] What is the Builder pattern used for?**

A) Sharing state between many small objects
B) Constructing complex objects step by step
C) Providing access to a remote object
D) Adding responsibilities to objects dynamically

Answer: B

**Q37. [Difficulty: 6.5] What is the N+1 query problem?**

A) Querying a table with N+1 columns
B) Having N+1 replicas of a database
C) Running one query to fetch a list and then one extra query for each item in it
D) Running N queries in parallel

Answer: C

**Q38. [Difficulty: 6.5] What does it mean for an API operation to be idempotent?**

A) The operation never modifies data
B) Repeating the same request has the same effect as sending it once
C) The operation always returns the same response time
D) The operation can be called only once

Answer: B

**Q39. [Difficulty: 6.5] What is a dead letter queue?**

A) A queue that holds messages that could not be processed successfully
B) A queue that deletes all messages immediately
C) A queue used only for emails
D) A queue for messages older than a year

Answer: A

**Q40. [Difficulty: 7] Which design pattern is typically used to implement undo and redo operations?**

A) Flyweight
B) Command
C) Prototype
D) Singleton

Answer: B

**Q41. [Difficulty: 7] What is the difference between optimistic and pessimistic locking?**

A) Optimistic locking checks for conflicts (e.g., a version number) at update time; pessimistic locking locks the data up front
B) Optimistic locking is used only in NoSQL databases
C) Optimistic locking locks rows first; pessimistic locking never locks
D) They are the same technique

Answer: A

**Q42. [Difficulty: 7] What is the purpose of a circuit breaker in a microservice architecture?**

A) To automatically scale servers
B) To balance load between replicas
C) To stop calling a failing service for a while and avoid cascading failures
D) To encrypt traffic between services

Answer: C

**Q43. [Difficulty: 7.5] What does CQRS stand for and what does it do?**

A) Concurrent Query Request Scheduling: schedules queries in parallel
B) Cache Query Response Storage: caches query results
C) Central Queue Retry System: retries failed messages
D) Command Query Responsibility Segregation: separates the models for reads and writes

Answer: D

**Q44. [Difficulty: 7.5] What is a Bloom filter?**

A) A structure that may give false negatives but never false positives
B) A sorted tree that stores unique keys
C) A probabilistic structure that may give false positives but never false negatives for membership
D) A cache that removes the oldest items first

Answer: C

**Q45. [Difficulty: 7.5] How does the Saga pattern differ from two-phase commit (2PC)?**

A) A saga uses a sequence of local transactions with compensating actions instead of a blocking global commit
B) A saga locks all databases until everything commits
C) A saga works only with a single database
D) A saga guarantees isolation across services

Answer: A

**Q46. [Difficulty: 8] In a 5-node Raft cluster, how many node failures can the cluster tolerate while still making progress?**

A) 3
B) 4
C) 2
D) 1

Answer: C

**Q47. [Difficulty: 8] What is a cache stampede (thundering herd on a cache)?**

A) The cache running out of memory
B) A cache returning stale data for a year
C) Two caches holding different values
D) Many requests miss the cache at once for an expired hot key and all hit the database

Answer: D

**Q48. [Difficulty: 8] What is the main risk of a write-behind (write-back) cache?**

A) Reads are always slower
B) Data may be lost if the cache fails before it is written to the database
C) The cache can never be updated
D) The database is written twice

Answer: B

**Q49. [Difficulty: 8.5] In Dynamo-style quorum systems with N replicas, read quorum R and write quorum W, which condition gives strong consistency?**

A) $W + R = 1$
B) $W + R < N$
C) $W + R > N$
D) $W = R = 1$

Answer: C

**Q50. [Difficulty: 8.5] What are vector clocks used for in distributed systems?**

A) Encrypting timestamps
B) Synchronizing server clocks exactly
C) Detecting causality and conflicting concurrent updates
D) Measuring network latency

Answer: C

**Q51. [Difficulty: 8.5] Which anomaly can still occur under the READ COMMITTED isolation level?**

A) Dirty read
B) Non-repeatable read
C) Reading uncommitted data
D) Lost power

Answer: B

**Q52. [Difficulty: 9] What is a hot partition (hot key) problem?**

A) One shard or key receives a disproportionately large share of the traffic
B) A database that has too many indexes
C) A server overheating
D) A partition that has no data

Answer: A

**Q53. [Difficulty: 9] In a news feed design, why is fan-out-on-write a problem for users with millions of followers?**

A) Followers cannot read the post
B) The post cannot be deleted
C) One post must be written into millions of follower feeds, which is very expensive
D) The post cannot be cached

Answer: C

**Q54. [Difficulty: 9] Which approach generates roughly time-sortable unique IDs across distributed servers without central coordination?**

A) Random 4-digit numbers
B) Auto-increment column in one database
C) Hashing the user's name
D) Snowflake-style IDs (timestamp + machine ID + sequence)

Answer: D

**Q55. [Difficulty: 9.5] What are LSM trees (used in Cassandra and RocksDB) optimized for?**

A) Reducing the number of disk reads for every query
B) High write throughput through sequential writes and background compaction
C) Storing data only in memory
D) Fast in-place updates of single rows

Answer: B

**Q56. [Difficulty: 9.5] What does the PACELC theorem add to the CAP theorem?**

A) It replaces availability with durability
B) Even without a partition, there is a trade-off between latency and consistency
C) It states partitions never happen
D) It adds a fourth guarantee called ordering

Answer: B

**Q57. [Difficulty: 9.5] What is hinted handoff in Dynamo-style databases?**

A) A node temporarily stores writes meant for an unavailable replica and delivers them when it returns
B) Sending hints to clients about slow nodes
C) Moving a shard to a new cluster
D) Handing the leader role to another node

Answer: A

**Q58. [Difficulty: 10] What are Merkle trees used for in distributed databases?**

A) Balancing load between clients
B) Encrypting replicated data
C) Sorting keys across shards
D) Efficiently detecting differences between replicas to repair them (anti-entropy)

Answer: D

**Q59. [Difficulty: 10] What is the dual-write problem and a common fix?**

A) Two replicas writing at once; fixed by deleting one
B) Writing twice to the same table; fixed with an index
C) Writing to a database and a message broker separately can leave them inconsistent; the transactional outbox pattern fixes it
D) Two users writing at once; fixed with caching

Answer: C

**Q60. [Difficulty: 10] What is HyperLogLog used for?**

A) Estimating the number of distinct elements using very little memory
B) Sorting very large files
C) Detecting deadlocks
D) Compressing logs

Answer: A